

NOVA PROGRAM

Nova High-Voltage Electrical Requirements

Domain: Electrical safety

Revision A | Synthetic extraction benchmark

2. Functional requirements

REQ-NVA-ELE-001 | Operating voltage window

The traction battery controller shall remain functional from 350 V DC through 820 V DC.

REQ-NVA-ELE-002 | Sleep current

After the wake line has remained low for 120 seconds, controller supply current shall not exceed 5 mA.

REQ-NVA-ELE-003 | Post-impact isolation

At 60 seconds after an impact event, the measured isolation resistance divided by the measured bus voltage shall be at least 500 ohm/V.

3. Verification-oriented requirements

Identifier	Requirement title	Normative requirement
REQ-NVA-ELE-004	Contactor opening	When a validated crash input is received, the positive and negative main contactors shall each reach the open state within 20 ms.
REQ-NVA-ELE-005	Precharge behavior	During each high-voltage start, the DC link shall reach at least 90 percent of pack voltage within 2.0 s without exceeding pack voltage by more than 10 V.

Applicability notes

Context	Interpretation
Unless a requirement states otherwise	Limits apply to the complete production-intent assembly.
Use of shall	The complete sentence is normative; table columns do not split obligations.

4. Normative installation figure

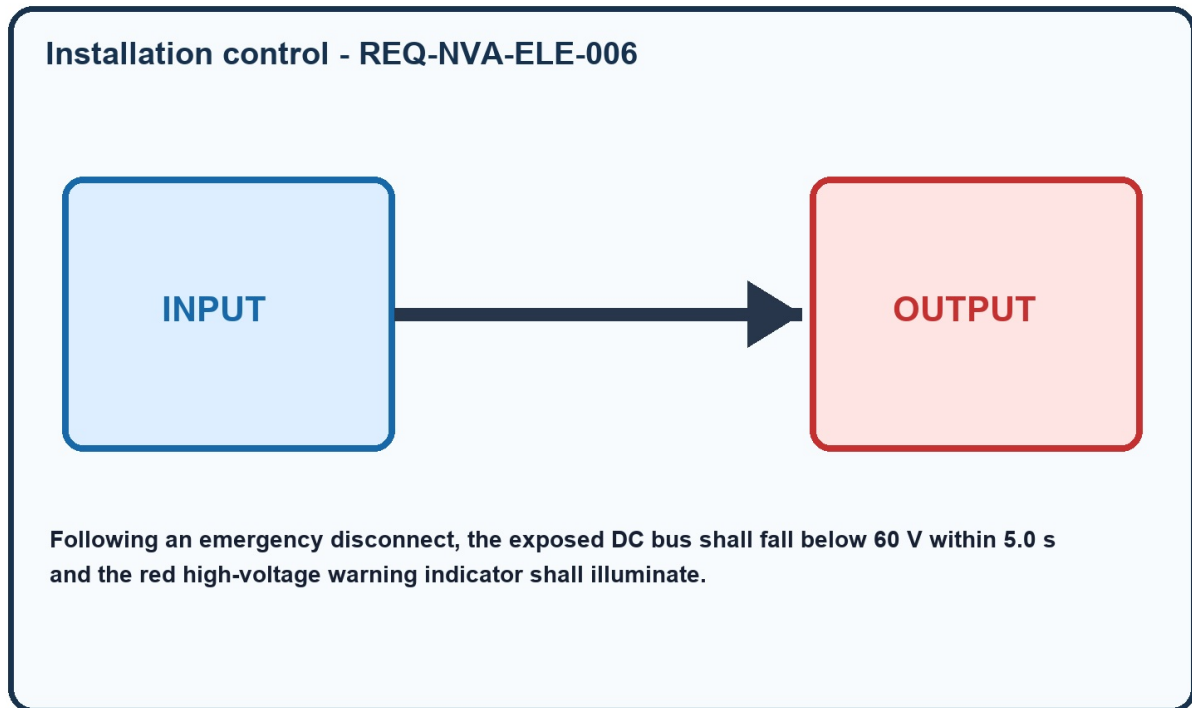


Figure 4-1. Colored geometry and embedded annotation are normative.

Engineering note

The figure annotation forms part of the controlled requirement baseline.

Color, direction, sequence, and geometric relationships must be interpreted visually.